

# FII/DII Flow Divergence as a Predictor of CAPM Pricing Error Direction

A null finding (corrected)

Dataset: Indian equity markets, 2015-01-01 to 2022-12-08 (training window)

Holdout: Aug 2025 - Apr 2026 (167 days), isolated and unused in training -- see Section 6

Signals tested: 2-of-3 concordance (flow divergence, PCR deviation, VIX implied-realized vol spread)

Run date: 2026-06-16

Supersedes: finding.zip (2026-04-07)

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*The earlier version was generated from a dataset that had the intended out-of-sample holdout merged into it two days later. Its reported statistics no longer match anything reproducible in this repo. Section 7 documents all corrections.*

## Abstract

We test whether daily FII/DII equity flow divergence predicts the direction of CAPM pricing errors in four Indian sector indices (Bank, IT, Pharma, Auto) over 1-, 3-, 5-, and 10-day forward horizons, using 1,883 days of real equity FII/DII flow data (2015-01-01 to 2022-12-08).

We find no predictive edge. Across 16 sector/horizon cells, none survive Bonferroni/BH correction for multiple comparisons. The robustness sweep shows significance in 5.6% of 160 parameter combinations, statistically indistinguishable from the 5% chance baseline.

We also test, with a reproducible and correctly-constructed script (1,750-day overlap, comparing flow-equivalent change rather than raw level), whether F&O; participant OI proxies equity cash flows. It does not. Sign agreement is 48.9%, indistinguishable from chance and stable across five sub-periods. This contradicts a prior claim that F&O; OI was "structurally inverted" -- that claim used only 100 days and a stock/flow construction mismatch that produced spurious anti-correlation.

## 1. Hypothesis

Research question: do days when FII and DII flows are simultaneously extreme and divergent coincide with a predictable CAPM mispricing direction in Indian sector indices?

The prior mechanism: when FII sells heavily while DII buys (or vice versa), one side is systematically wrong about fair value. If large enough, that disagreement should bias next-day sector returns relative to a rolling-beta CAPM prediction.

## 2. Data and methodology

### 2.1 Market data

yfinance, Nifty 50 (^NSEI), India VIX (^INDIAVIX), 4 sector indices, 2015-01-01 to 2022-12-08.

### 2.2 Institutional flow data

data/raw/fii\_dii.csv -- 1,883 rows, 2015-01-01 to 2022-12-08, INR Crores. This is the only FII/DII data used for training and the main test. The Aug 2025-Apr 2026 segment previously merged into this file has been moved to data/holdout/fii\_dii\_holdout.csv and excluded from config.py's date range, so it cannot be silently re-absorbed into a future run.

## 2.3 Signal construction

Unchanged from the original design: flow divergence (FII-DII, 252-day percentile rank), PCR deviation (z-score vs 30-day baseline, 252-day percentile rank), vol spread (VIX - 20-day realized vol, 252-day percentile rank). All three use trailing-only windows (.rolling(window), no centering). No look-ahead bias in signal construction.

## 2.4 CAPM pricing error

Rolling 60-day OLS per sector, beta estimated strictly on the prior 60 days (iloc[i-60:i], excludes day i). Verified: no leakage.

## 2.5 Anomaly detection

2-of-3 signals simultaneously in top/bottom 10th percentile (unchanged).

## 2.6 Statistical test -- corrected for autocorrelation

This is the main methodological fix in this revision. The original sign test treated every flagged day as an independent Bernoulli trial. But flagged days cluster: in the training data, 65% of consecutive flagged-day pairs fall within 3 trading days of each other, so their forward-error windows overlap and their signs are correlated, not i.i.d. That inflates apparent significance.

Fix (test\_forward\_prediction\_episodes in test\_prediction.py): consecutive flagged days are collapsed into one episode, evaluated from the day after the episode ends. Episodes whose forward window would still overlap the next episode's trigger are dropped rather than counted, so retained trials are genuinely non-overlapping. This trades sample size (n drops from ~170 to ~55-105 depending on horizon) for validity. Both the naive and corrected numbers appear in results/main\_results.csv for transparency. The corrected column is the one to trust.

# 3. Results

## 3.1 Main test (corrected, non-overlapping episodes)

| Sector       | 1d (n, hit, p)       | 3d                         | 5d                  | 10d                  |
|--------------|----------------------|----------------------------|---------------------|----------------------|
| NIFTY_BANK   | n=105, 42.9%, p=0.17 | n=86, 38.4%, p=0.040 (sig) | n=77, 49.4%, p=1.0  | n=55, 47.3%, p=0.79  |
| NIFTY_IT     | n=105, 49.5%, p=1.0  | n=86, 58.1%, p=0.16        | n=77, 51.9%, p=0.82 | n=55, 47.3%, p=0.79  |
| NIFTY_PHARMA | n=105, 51.4%, p=0.85 | n=86, 50.0%, p=1.0         | n=77, 44.2%, p=0.36 | n=55, 36.4%, p=0.058 |
| NIFTY_AUTO   | n=105, 46.7%, p=0.56 | n=86, 51.2%, p=0.91        | n=77, 50.6%, p=1.0  | n=55, 45.5%, p=0.59  |

1 of 16 cells significant at alpha=0.05 -- at or below the ~0.8 expected by chance with 16 comparisons, before any Bonferroni correction. NIFTY\_BANK's 3-day hit rate is below 50%, so it is anti-predictive of the naive directional hypothesis.

## 3.2 Walk-forward (3-day forward, corrected test, 3 full 2-year windows + 1 partial year)

| Period              | BANK           | IT             | PHARMA | AUTO  |
|---------------------|----------------|----------------|--------|-------|
| 2016-2017 (n=18)    | 38.9%          | 38.9%          | 66.7%  | 55.6% |
| 2018-2019 (n=31)    | 41.9%          | 67.7%          | 38.7%  | 45.2% |
| 2020-2021 (n=26)    | 23.1%, p=0.009 | 73.1%, p=0.029 | 61.5%  | 46.2% |
| 2022 partial (n=10) | 70.0%          | 20.0%          | 30.0%  | 70.0% |

NIFTY\_BANK's hit rate runs from 23% to 70% across periods. There is no stable relationship. The 2020-2021 significance for both BANK and IT looks COVID-regime-specific, not a generalizable edge. The 2022 row is a partial year (Jan-Dec 8 only, n=10); treat it as low-power.

Note: the previous "2024-2025" walk-forward row has been removed. The underlying data has zero rows for 2023-2024, and what remained (2025-26) was the leaked holdout, now correctly excluded.

### 3.3 Signal decomposition (3-day forward, corrected test)

| Signal                  | BANK  | IT              | PHARMA | AUTO  |
|-------------------------|-------|-----------------|--------|-------|
| flow_div alone (n=165)  | 50.9% | 47.9%           | 48.5%  | 52.1% |
| pcr_dev alone (n=106)   | 44.3% | 60.4% (p=0.041) | 46.2%  | 49.1% |
| vol_spread alone (n=61) | 39.3% | 50.8%           | 44.3%  | 47.5% |

No signal in isolation shows a reliable edge. The one marginal hit for PCR/IT is not replicated elsewhere.

### 3.4 Robustness sweep

9 of 160 parameter combinations (5.6%) show  $p < 0.05$ . That is consistent with the ~5% expected under the null, and -- unlike the previous write-up -- this number is reproducible directly from results/robustness.csv. Eight of the 9 significant combinations are NIFTY\_BANK at 3-day forward. The recurring pattern (hit rate 29%-41%, consistently below 50%) may be idiosyncratic to NIFTY\_BANK at that horizon, but it does not survive walk-forward (direction flips by period) and is not strong enough to call a real edge.

## 4. F&O; OI proxy check -- now reproducible, and the original claim doesn't hold

This section was revised twice. The original (finding.zip) cited a Spearman  $r = -0.019$  / 28%-sign-agreement claim with no script or data backing it anywhere in the repo. A first fix (validate\_oi\_proxy.py rev.1) made the number reproducible but only checked a 100-day window, comparing the F&O; OI level (a stock, cumulative outstanding position) against equity flow (a daily transaction). That got  $r = -0.011$ , 36% sign agreement, which appeared to confirm the original "structurally inverted" claim.

That confirmation does not hold under a larger, correctly-constructed test. Two problems were identified independently:

1. Too little data. The training equity data (2015-2022) overlaps the OI archive for 1,750 days, not just the 100-day holdout slice.
2. Stock vs. flow mismatch. F&O; net OI is a level; equity net flow is a daily transaction amount. Comparing them directly conflates "is the outstanding position growing in the same direction as today's

equity buying" with "is today's F&O; activity in the same direction as today's equity activity." The economically defensible comparison uses the day-over-day change in F&O; net OI (the flow-equivalent), not the level.

| Construction                                      | n     | Spearman r      | Sign agreement |
|---------------------------------------------------|-------|-----------------|----------------|
| LEVEL (as originally tested)                      | 1,750 | -0.017 (p=0.47) | 40.5%          |
| CHANGE (flow-equivalent, defensible construction) | 1,749 | -0.002 (p=0.95) | 48.9%          |

Broken out by sub-period (CHANGE construction):

| Period       | n   | r      | Sign agreement |
|--------------|-----|--------|----------------|
| 2015-2016    | 425 | +0.057 | 51.5%          |
| 2017-2018    | 430 | -0.043 | 45.3%          |
| 2019-2020    | 438 | -0.012 | 51.1%          |
| 2021-2022    | 356 | +0.017 | 47.5%          |
| Aug-Dec 2025 | 100 | +0.012 | 49.0%          |

F&O; participant OI carries no detectable relationship to equity flow direction. Sign agreement is roughly chance-level, stable at 45-52% across five multi-year periods. The original "structurally inverted" or "anti-correlated" framing is not supported. If that hedging mechanism reliably drove a sign inversion, sign agreement should sit consistently below 50% across periods. Instead it straddles 50% with no consistent direction.

The practical conclusion is the same (don't use F&O; OI level as an equity-flow proxy -- it doesn't work) but the reason is different and weaker than originally claimed. It's not "actively misleading," it's just uninformative. That distinction matters if anyone downstream were to consider inverting the F&O; signal as a contrarian proxy. That would not work either, since there's no stable relationship in either direction.

## 5. Holdout (Aug 2025 - Apr 2026) -- could not be validated

We ran a genuine holdout check (holdout\_check.py) rather than skip it. Result: it doesn't work, and shouldn't be presented as if it does.

The main methodology needs a 252-day rolling window. The holdout is only 167 days, separated from training by a 2.5-year gap with zero FII/DII coverage (2023-2024). There is no way to compute a "trailing year" feature for the holdout that means the same thing it meant during training. A reduced 60-day-window variant, confined entirely to holdout data, was run as a fallback. It produced only 1 flagged day across the whole holdout period, which is insufficient for any statistical test (see results/holdout\_check.csv).

A valid out-of-sample validation is not currently possible with the data available. Closing this gap requires backfilling FII/DII data for 2023-2024, not a code fix.

### 5.5 Extended validity diagnostics (validity\_checks.py)

Four additional checks, run after external review of the corrections above: whether episode-correction alone was enough, whether 16 comparisons needed an explicit multiple-testing correction, and whether

hit-rate significance corresponds to anything economically meaningful.

### Multiple testing correction

Bonferroni and Benjamini-Hochberg FDR applied to the 16 main sector/horizon p-values (now in columns p\_bonferroni, p\_bh\_fdr in results/main\_results.csv).

Bonferroni threshold:  $0.05/16 = 0.0031$

NIFTY\_BANK 3-day ( $p=0.0399$ ) does not clear it.

0 of 16 cells significant after either correction. This is the strongest statement of the null finding in this document: even the one cell that survived episode-correction does not survive accounting for the fact that 16 hypotheses were tested.

### Permutation test (5,000 shuffles, preserves flag run-length structure)

| Sector             | 3d hit rate | p (permutation)               |
|--------------------|-------------|-------------------------------|
| NIFTY_BANK         | 38.4%       | 0.038                         |
| All other 15 cells | --          | $p > 0.07$ , mostly $p > 0.4$ |

The permutation p for NIFTY\_BANK (0.038) closely tracks the parametric binomial p (0.040), confirming the binomial approximation was not the source of the apparent significance. (Full table: results/permutation\_test.csv.)

### Block bootstrap (20-day blocks, 2000 resamples) -- NIFTY\_BANK 3-day only

| Metric                                                        | Value |
|---------------------------------------------------------------|-------|
| Naive n                                                       | 170   |
| Variance inflation factor (block vs. naive binomial variance) | 1.41  |
| Effective n                                                   | ~120  |
| Block-bootstrap p-value                                       | 0.027 |

Regime clustering inflates the variance by ~41%, but this alone does not explain away the NIFTY\_BANK 3-day result. The multiple-testing correction above is what kills it. Full output: results/block\_bootstrap\_nifty\_bank\_3d.csv.

### Economic significance (mean forward return and Sharpe-like ratio, flagged vs. unconditional)

| Sector       | 3d mean return, flagged | 3d mean return, unconditional | Sharpe-like, flagged |
|--------------|-------------------------|-------------------------------|----------------------|
| NIFTY_BANK   | -0.032%                 | +0.066%                       | -0.024               |
| NIFTY_IT     | +0.161%                 | +0.073%                       | +0.188               |
| NIFTY_PHARMA | +0.152%                 | +0.023%                       | +0.134               |
| NIFTY_AUTO   | +0.067%                 | +0.042%                       | +0.054               |

All magnitudes are small (Sharpe-like ratios well under 0.3). None of IT/PHARMA/AUTO's flagged-day deltas correspond to a statistically significant hit rate in Section 3.1. Hit-rate significance and economic magnitude are different questions. Neither would plausibly survive transaction costs even where the sign looks favorable. Full table: results/economic\_significance.csv.

Net effect of this diagnostic pass: the null finding is airtight, not just probable.

## 6. Limitations

The non-independence correction trades power for validity. The corrected test has roughly half the naive  $n$ . A real effect of moderate size could be missed. Section 5.5 shows the residual autocorrelation effect is modest ( $VIF \sim 1.4$ ), so this is not a large hidden cost.

The sign test only captures direction, not magnitude. This is addressed partially in Section 5.5's economic-significance check, which shows the magnitudes involved are small regardless.

The 2023-2024 data gap (see Section 5) blocks any holdout validation under the current design.

PCR data is included in the 2-of-3 concordance filter in this revision (the previous version ran 2-of-2 without it). All results above reflect 3-signal concordance.

The equity flow source is single and has not been cross-validated against a second provider.

Data provenance is now hashed but not fully independently verified. `generate_manifest.py` produces `data/MANIFEST.md` with SHA256/size/mtime for every raw file, and explicitly labels each file's source claim as either verified (yfinance/NSE data this repo's own code re-fetches live) or declared (the Kaggle FII/DII CSV and the NSE F&O; OI archive, whose provenance rests on a written claim that nothing here independently re-validates against the original source). This closes the "could be silently swapped" gap but does not prove the Kaggle data is what it claims to be. That would require re-downloading from Kaggle and diffing, which hasn't been done.

CAPM may be a noisy benchmark for this question, and has not been tested against alternatives. A real flow effect could exist and still not register against rolling-beta CAPM pricing-error sign. It might instead show up in residual returns, sector-relative returns, or factor-neutral returns (Fama-French-style). This project rules out flow divergence as a predictor of CAPM error sign. It does not rule out flows mattering for other target variables. Re-testing against a different target would need either factor data (not currently in this repo) or a redesigned signal-construction step.

## 7. Corrections from the previous version (finding.zip, 2026-04-07)

### 1. Holdout leakage fixed

`data/raw/fii_dii.csv` had the Aug2025-Apr2026 holdout merged into it by the time of the previous run. It has been split back into `data/raw/fii_dii.csv` (train, up to 2022-12-08) and `data/holdout/fii_dii_holdout.csv`, and `config.py`'s `END_DATE` now hard-bounds the training window.

### 2. Robustness sweep number corrected

Previous claim: "13/160 (8.1%)". That number didn't match `results/robustness.csv` even at the time (actual:  $34/160 = 21.25\%$  on the leaked dataset). Current, correct, reproducible number: 9/160 (5.6%) on clean data.

### 3. Sign test independence fixed

The previous version used every flagged day as an independent trial despite heavy autocorrelation. The corrected episode-based test is now the default; naive numbers are kept alongside for comparison only.

#### **4. F&O; proxy claim made reproducible, then found not to replicate**

First pass (validate\_oi\_proxy.py rev.1) made the number reproducible but only checked a 100-day window using a level-vs-flow construction mismatch, and got a result that appeared to confirm the original "structurally inverted" claim. A second pass (rev.2, prompted by review questioning the construction and overlap period) used the full 1,750-day overlap and the correct flow-equivalent (day-over-day OI change) construction, and found no relationship at all (48.9% sign agreement, chance level, stable across 5 sub-periods). See Section 4.

#### **5. Walk-forward periods fixed**

Removed the "2024-2025" period (no longer meaningful given the data gap and the holdout exclusion); replaced with the real partial-year 2022 window.

#### **6. Holdout validation attempted honestly**

Rather than silently dropping the holdout or faking a result, holdout\_check.py now documents exactly why it cannot currently produce a valid result.

#### **7. Extended validity diagnostics added (validity\_checks.py, Section 5.5)**

In response to external review: multiple-testing correction (Bonferroni/BH), a distribution-free permutation test, a block bootstrap to quantify how much regime-clustering inflates significance, and economic-significance metrics (mean return, Sharpe-like ratio) since hit-rate alone doesn't establish exploitability.

#### **8. Data provenance manifest added**

In response to further review: generate\_manifest.py now produces data/MANIFEST.md with SHA256 and verification status (independently re-fetchable vs. merely declared) for every raw data file.

### **8. Conclusions**

#### **1. The null finding holds, and is now airtight**

FII/DII flow divergence (2-of-3 concordance with PCR and vol spread) does not predict CAPM error direction in Indian equities, 2015-2022. This holds under the corrected non-overlapping-episode sign test, a distribution-free permutation test, and -- after Bonferroni/BH correction for the 16 sector/horizon comparisons -- 0 of 16 cells are significant.

#### **2. NIFTY\_BANK at 3-day forward was the one interesting cell**

It survived episode-correction ( $p=0.040$ ) and the permutation test ( $p=0.038$ ), with a block-bootstrap effective  $n$  of  $\sim 120$  (down from 170, but not collapsed). It is killed by the multiple-testing correction, and independently undermined by walk-forward instability (23%-70% hit rate range across regimes) and a negative, noise-level economic effect (Sharpe-like  $-0.024$  at 3-day). Not exploitable by any measure applied here.

#### **3. F&O; participant OI is uninformative about equity flow direction**

The original (and first-revision) claim that it was "structurally inverted" or "anti-correlated" does not survive a properly-constructed, full-overlap test: sign agreement is 48.9%, chance-level, stable across five sub-periods spanning 2015-2025. The practical conclusion is the same (don't use it as a proxy) but

the mechanism claimed in the original write-up (FII hedging causes a sign inversion) is not supported. It's noise, not a backwards signal. This is the clearest example in this project of why "reproducible" and "correct" are different bars: rev.1 was fully reproducible and still wrong, because the construction (level vs. flow, 100 days vs. 1,750) was the actual problem.

#### 4. A valid out-of-sample test is not currently possible

Backfilling the 2023-2024 FII/DII gap is the prerequisite. This is now stated directly rather than implied as a vague next step while the holdout was quietly already consumed.

#### 5. Remaining honest gaps, in order of effort to close

(a) Data provenance: data/MANIFEST.md hashes everything and labels what's independently re-verifiable vs. merely declared, but the Kaggle source itself is still unverified against its origin. (b) The 2023-2024 gap: needs new data, not more analysis. (c) The CAPM target-variable choice: the most interesting open question, which requires re-testing against residual, sector-relative, or factor-neutral returns -- a genuine scope expansion, not a bug fix.

### Appendix: pipeline summary

| Layer | File                 | Description                                                                           |
|-------|----------------------|---------------------------------------------------------------------------------------|
| L0    | data_pipeline.py     | Market data (yfinance) + FII/DII CSV ingestion + alignment                            |
| L1    | signals.py           | Flow divergence, PCR deviation, VIX vol spread (252-day percentile)                   |
| L2    | capm.py              | Rolling 60-day OLS CAPM per sector; pricing error computation                         |
| L3    | detector.py          | 2-of-N concordance flag at configurable percentile threshold                          |
| L4    | test_prediction.py   | Episode-corrected sign test + walk-forward + decomposition + robustness sweep         |
| L5    | validity_checks.py   | Bonferroni/BH correction + permutation test + block bootstrap + economic significance |
| --    | validate_oi_proxy.py | Reproducible F&O-OI-vs-equity-flow comparison                                         |
| --    | holdout_check.py     | Honest holdout attempt + documented failure mode                                      |
| --    | generate_manifest.py | SHA256 + source-verification-status manifest for all raw data files                   |
| --    | run.py               | End-to-end orchestrator (main test only; holdout/proxy/validity run separately)       |
| --    | fetch_archives.py    | NSE Archives PCR fetch (F&O OI to PCR only; FII/DII export disabled)                  |

Reproducibility: `python3 run.py -> python3 validity_checks.py` for the main result and its diagnostics; `python3 validate_oi_proxy.py` and `python3 holdout_check.py` for the two ancillary checks. All numbers in this document were generated by these exact scripts on 2026-06-16.